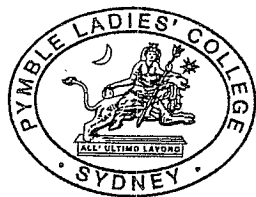


Name: _____

Teacher: _____



PYMBLE LADIES' COLLEGE

MATHEMATICS

TRIAL HSC EXAMINATION

2006

Reading Time: 5 minutes
Working Time: 3 hours

Instructions to students:

- Write using blue or black biro.
- All questions may be attempted.
- Diagrams are not to scale.
- All necessary working should be shown in every question.
- Your name and your teacher's name may be written before you begin the assessment.
- Start each question in a new booklet.
- Marks may be deducted for careless or untidy work.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

Question 1 (12 Marks) Use a SEPARATE writing booklet

Marks

- (a) Evaluate $\frac{4.26 + 3.81}{3\sqrt{6.27}}$. Give the answer correct to three significant figures.

2

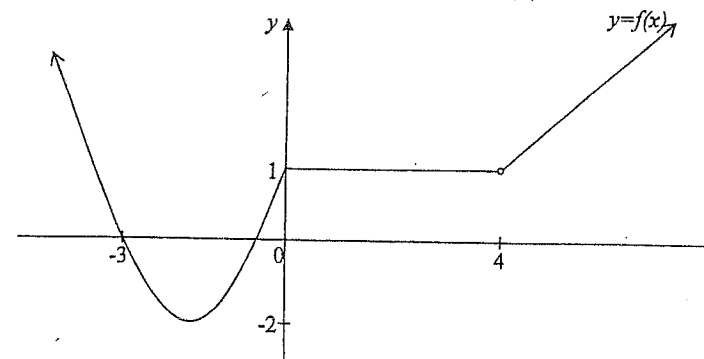
- (b) Factorise $x^3 - 8$.

1

- (c) Find the values of a and b such that $\frac{8}{\sqrt{5}-3} = a - \sqrt{b}$.

3

- (d) The diagram below represents the graph of $y = f(x)$.



State its domain and range.

2

- (e) Find the values of x for which $|5 - 2x| \geq 1$.

2

- (f) Differentiate $\frac{\sqrt{x}}{4}$ with respect to x .

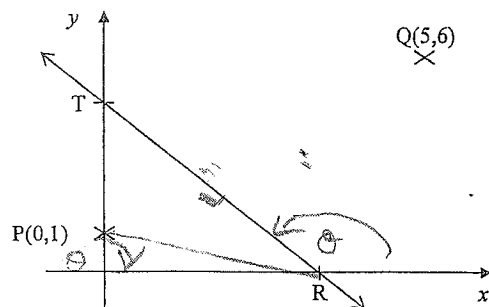
2

Question 2 (12 Marks) Use a SEPARATE writing booklet

Marks

In the diagram, the points P and Q have coordinates (0,1) and (5,6) respectively. The line through T and R has equation $y = \frac{5-2x}{2}$.

Copy the diagram onto your answer booklet.



(i) Find the size of the angle which the line PQ makes with the positive direction of the x-axis.

2

(ii) Show that the equation of the line PQ is $x - y + 1 = 0$.

1

(iii) Given M is the point where the line PQ intersects the line RT, find the coordinates of M.

2

(iv) Find the perpendicular distance from P to the line RT.

2

(v) If $\angle PMR$ is a right-angle, then find the area of $\triangle PRM$.

3

(vi) On your diagram shade the region which satisfies $x - y + 1 \geq 0$, $2x + 2y \leq 5$ and $y \geq 0$ simultaneously.

2

Question 3 (12 Marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate $2x \tan x$ with respect to x .

2

(b) Given $f(x) = x^3 - 4x^{-1}$, find the value of $f'(\sqrt{2})$.

2

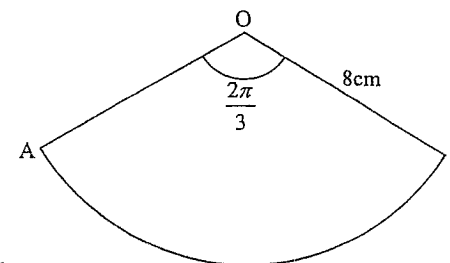
(c) Evaluate $\int_0^{\frac{\pi}{9}} \cos 3x \, dx$.

2

(d) Find $\int \frac{3}{1+2x} \, dx$.

2

(e) A cone is formed by folding the sector ABO so that the edges OA and OB coincide.



Find:

(i) the exact area of the sector ABO.

1

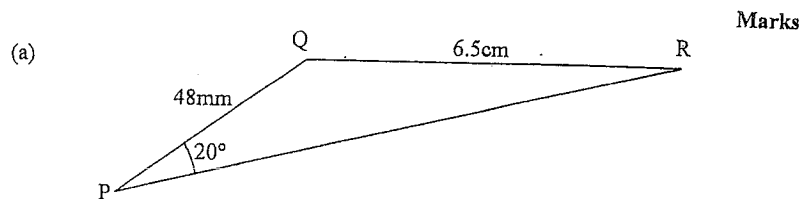
(ii) the exact length of the arc AB.

1

(iii) the radius of the base of the cone formed.

2

Question 4 (12 Marks) Use a SEPARATE writing booklet



PQR is a triangle with $PQ = 48$ mm, $QR = 6.5$ cm and $\angle QPR = 20^\circ$. Find the size of $\angle PQR$ correct to the nearest degree.

- (b) Consider the function $f(x) = -x^5 + 15x^3$.
- (i) Find the coordinates of the stationary points of the curve $y = f(x)$ and determine their nature. 4
- (ii) Sketch the curve showing all important features including the x -intercepts and points of inflexion. 4
- (iii) State the values of x for which the curve $y = f(x)$ is concave down. 1

Question 5 (12 Marks) Use a SEPARATE writing booklet

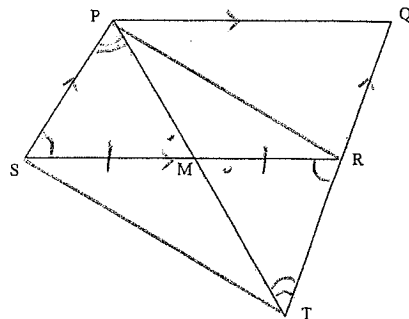
- (a) Consider the parabola $(y-4)^2 = 8(x+2)$.
- (i) Write down the coordinates of the vertex. 1
- (ii) Find the focus. 2
- (iii) Find the y -intercepts. 2
- (b) If α and β are the roots of the quadratic equation $3x^2 - 4x - 1 = 0$, find the value of:
- (i) $\alpha + \beta$. 1
- (ii) $\alpha\beta$. 1
- (iii) $\alpha^2 + \beta^2$. 2
- (c) Given the sequence $\ln 4, \ln 16, \ln 64, \dots$
- (i) Show that it is arithmetic. 1
- (ii) Hence, find the sum of the first 20 terms in exact form. 2

Question 6 (12 Marks) Use a SEPARATE writing booklet

Marks

- (a) The velocity of a particle v cm/s moving in a straight line is given by $v = 1 + 2t - 3t^2$.
- If the initial displacement is 3 cm to the right of 0, calculate the displacement after 2 seconds. 2
 - When is the particle at rest? 2
 - How far does the particle travel in the third second? 2
 - Describe the motion of the particle. 1

(b)



PQRS is a parallelogram. M is the midpoint of SR.

PM produced meets QR produced at T.

- Prove that $\triangle PMS \cong \triangle TMR$. 3
- Prove that PRTS is a parallelogram. 2

Question 7 (12 Marks) Use a SEPARATE writing booklet

Marks

- Sketch the graph of $y = 3 \sin 2x$ for $0 \leq x \leq 2\pi$. 2
 - What is the period of the curve? 1
 - State the amplitude. 1
 - Find the area between the curve and the x -axis if $\frac{\pi}{4} \leq x \leq \frac{\pi}{2}$. 3
- Copy and complete the table of values for $y = 4^x$. 1

x	-1	0	1
y			

- Hence, using these three values and the trapezoidal rule, find an approximation for $\int_{-1}^1 4^x dx$. 2
- (α) Find the derivative of 4^x with respect to x . 1



Hence, or otherwise, find the exact value of $\int_{-1}^1 4^x dx$. 1

Question 8 (12 Marks) Use a SEPARATE writing booklet



- (a) Find the coordinates of the point on the curve $y = e^{3x}$ where the tangent is perpendicular to the line $y = 4 - \frac{x}{6}$.

Marks

3

- (b) (i) Show that $\sin^2 x \cos x = \cos x - \cos^3 x$.

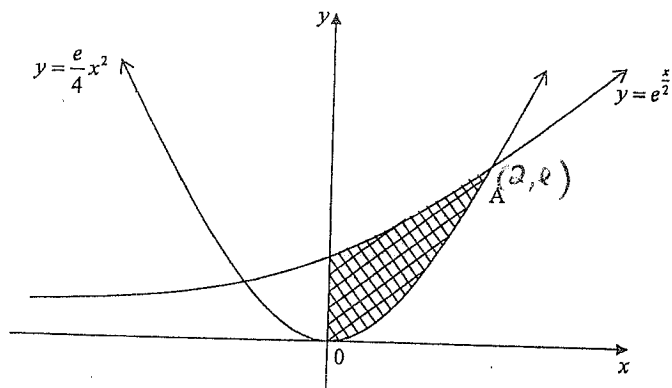
1



- (ii) Hence, find $\frac{d}{dx} \left(\sin x - \frac{1}{3} \sin^3 x \right)$.

3

(c)



The diagram above is of the exponential curve $y = e^{\frac{x}{2}}$ and the parabola $y = \frac{e}{4}x^2$. The point A is the first point where the two graphs meet on the right hand side of the y-axis.

- (i) Show that A is the point (2, e).

2

- (ii) Show that the shaded area is $\frac{4e}{3} - 2$ units².

3

Question 9 (12 Marks) Use a SEPARATE writing booklet

Marks

- (a) The rate of increase of a population $P(t)$ of people in a certain city is governed by the equation $\frac{dP}{dt} = kP$ where k is a constant and t is the time in years. The population of the city doubles every twenty years.

- (i) Show that $k = \frac{1}{20} \ln 2$.

2

- (ii) In which year will the city reach a population three times that which it had at the beginning of 2006?

2

- (iii) If at the beginning of 2010 the population is 20 million, what will be the population at the beginning of the year 2060?

2

- (b) Sarah wishes to buy a car. She has worked out that she can afford repayments of \$400 a month for 5 years.

The interest rate on offer is 24% pa (reducible) calculated monthly.

Let A_n be the amount owing after n months based on a monthly repayment of \$400 and P being the amount borrowed.

- (i) Give an expression for A_2 .

1

- (ii) Show that $A_n = 1.02^n P - 20000(1.02^n - 1)$.

3

- (iii) Hence, determine how much money Sarah is able to borrow?

2

Question 10 (12 Marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate $\ln(x + \sqrt{x^2 + a^2})$ and hence find $\int \frac{1}{2\sqrt{x^2 + a^2}} dx$. 3

(b) The region bounded by the curve $y = \ln x$, the axes and the line $y = \ln 2$, is rotated about the y -axis. 3

Find the volume of the solid formed.

(c) ABCDE is a pentagon of fixed perimeter P cm. Its shape is such that ABE is an equilateral triangle and BCDE is a rectangle.

If the length AB is x cm:

(i) Show that the length BC is $\frac{P-3x}{2}$ cm. 1

(ii) Show that the area of the pentagon is given by 2
 $A = \frac{1}{4} [2Px - (6 - \sqrt{3})x^2]$.

(iii) Find the value of $\frac{P}{x}$ for which the area of the pentagon is a 3
maximum.

End of paper

MARKING GUIDELINES

Q1

a) $\frac{4.26 + 3.81}{3\sqrt{6.27}}$
 $= 1.07428 \dots$
 $= 1.07 \text{ (3 s.f.)}$

(2)

b) $x^3 - 8$
 $= (x - 2)(x^2 + 2x + 4)$ 1 R/W (1)

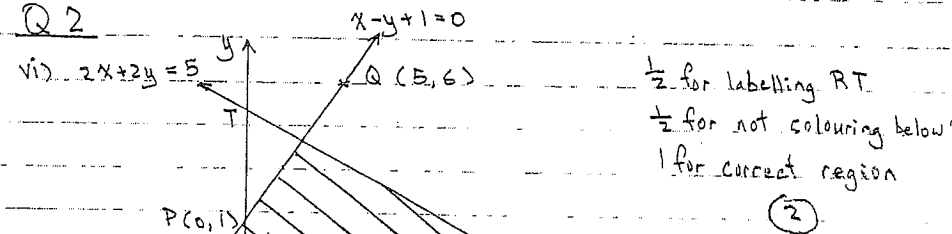
c) $\frac{8}{\sqrt{5} - 3}$
 $= \frac{8}{\sqrt{5} - 3} \times \frac{\sqrt{5} + 3}{\sqrt{5} + 3} \quad \frac{1}{2}$
 $= \frac{8(\sqrt{5} + 3)}{-4} \quad \frac{1}{2}$
 $= -2(\sqrt{5} + 3)$
 $= -6 - 2\sqrt{5} = -6 - \sqrt{20} \quad \frac{1}{2}$
 $\therefore a = -6 \frac{1}{2} \text{ and } b = 20 \frac{1}{2}$ (3)

d) Domain: all real x ; $x \neq 4$ 1 R/W
 Range: $y \geq -2$ 1 R/W (2)

e) $|5 - 2x| \geq 1$
 $5 - 2x \geq 1 \quad \text{OR} \quad -5 + 2x \geq 1 \quad \frac{1}{2}$
 $-2x \geq -4 \quad \frac{1}{2} \quad 2x \geq 6$
 $x \leq 2 \quad \frac{1}{2} \quad x \geq 3 \quad \frac{1}{2}$ (2)

f) $\frac{d}{dx} \left(\frac{\sqrt{x}}{4} \right)$
 $= \frac{d}{dx} \left(\frac{1}{4} x^{\frac{1}{2}} \right) \quad \frac{1}{2}$
 $= \frac{1}{4} \left(\frac{1}{2} x^{-\frac{1}{2}} \right) \quad \frac{1}{2} + \frac{1}{2}$
 $= \frac{1}{8} x^{-\frac{1}{2}} \quad \frac{1}{2}$
 $= \frac{1}{8\sqrt{x}}$ (2)

Q2



ii) $m_{PQ} = \frac{6-1}{5-0} = 1$
 Angle line PQ with x -axis = 45° 1 (2)

iii) Eq. of line PQ
 $\Rightarrow 1 = \frac{y-1}{x-0}$ 1 R/W
 $x = y - 1$
 $x - y + 1 = 0$ (1)

iii) $\begin{cases} x - y + 1 = 0 \\ 5 - 2x \end{cases}$
 $x - \left(\frac{5}{2} - x \right) + 1 = 0 \quad \frac{1}{2}$
 $2x = \frac{3}{2}$
 $x = \frac{3}{4}$ $\frac{1}{2}$
 $y = 1 \frac{3}{4}$ $\frac{1}{2}$
 $M = \left(\frac{3}{4}, 1 \frac{3}{4} \right)$ (2)

iv) Eq. of RT $\Rightarrow 2y = 5 - 2x$
 $2x + 2y - 5 = 0 \quad \frac{1}{2}$
 Perpendicular distance from P to RT
 $= \frac{|2(0) + 2(1) - 5|}{\sqrt{2^2 + 2^2}} \quad 1$
 $= \frac{3}{2\sqrt{2}} \quad \frac{1}{2}$
 $= \frac{3\sqrt{2}}{4}$ (2)

$$1) 2x + 2y - 5 = 0$$

$$y = 0 \Rightarrow 2x = 5$$

$$x = \frac{5}{2}$$

$$R(\frac{5}{2}, 0)$$

$$\begin{aligned} \text{Length of RM} &= \sqrt{(\frac{5}{2} - \frac{3}{4})^2 + (0 - \frac{3}{4})^2} \\ &= \frac{1\frac{3}{4} \times \sqrt{2}}{4} \\ &= \frac{7\sqrt{2}}{4} \end{aligned}$$

i.e. Area of $\triangle PRM$

$$= \frac{1}{2} \times \frac{7\sqrt{2}}{4} \times \frac{3}{2\sqrt{2}}$$

$$= \frac{1\frac{5}{16}}{\text{sq. units}}$$

Q3

$$a) \frac{d}{dx} (2x \tan x)$$

$$= 2 \tan x + 2x \sec^2 x$$

$$\text{let } u = 2x \\ \frac{du}{dx} = 2$$

$$v = \tan x \\ \frac{dv}{dx} = \sec^2 x$$

$$b) f(x) = x^3 - 4x^{-1}$$

$$f'(x) = 3x^2 + 4x^{-2}$$

$$f'(\sqrt{2}) = 3(\sqrt{2})^2 + 4(\sqrt{2})^{-2}$$

$$= 6 + 2$$

$$= 8$$

$$c) \int_0^{\frac{\pi}{3}} \cos 3x \, dx$$

$$= \frac{1}{3} \sin 3x \Big|_0^{\frac{\pi}{3}}$$

$$= \frac{1}{3} \sin \frac{\pi}{3} - \frac{1}{3} \sin 0$$

$$= \frac{\sqrt{3}}{6}$$

$$d) \int \frac{3}{1+2x} \, dx$$

$$= \frac{3}{2} \ln(1+2x) + C$$

e) i) Area of sector ABO

$$= \frac{1}{2} \times 8^2 \times \frac{2\pi}{3}$$

$$= \frac{64\pi}{3} \text{ cm}^2$$

$$\text{ii) Arc AB} = 8 \times \frac{2\pi}{3}$$

$$= \frac{16\pi}{3} \text{ cm}$$

iii) Circumference of base = length of arc AB

$$2\pi r = \frac{16\pi}{3}$$

$$r = \frac{8}{3}$$

i.e. Radius of base of cone

$$= 2\frac{2}{3} \text{ cm}$$

Q4

$$a) \frac{\sin \theta}{4.8} = \frac{\sin 20^\circ}{6.5}$$

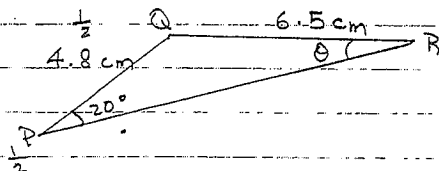
$$\sin \theta = \frac{\sin 20^\circ}{6.5} \times 4.8$$

$$\theta = 14.629^\circ \dots$$

$$\angle PQR = 180^\circ - 20^\circ - 14.629^\circ \dots$$

$$= 145^\circ 22'$$

$$= 145^\circ$$



(3)

$$b) f(x) = -x^5 + 15x^3$$

$$i) f'(x) = -5x^4 + 45x^2 = 0$$

$$= -5x^2(x^2 - 9) = 0$$

$$x = 0 \text{ or } x = 3 \text{ or } x = -3$$

$$y = 0 \text{ or } y = 162 \text{ or } y = -162$$

$$f''(x) = -20x^3 + 90x$$

$$f''(0) = 0$$

$$f''(3) = -20 \times 3^3 + 90 \times 3 < 0$$

$$f''(-3) = -20(-3)^3 + 90(-3) > 0$$

$\therefore (0, 0)$ is a horizontal point of inflexion

$(3, 162)$ is a max. turning point

and $(-3, -162)$ is a min. turning point.

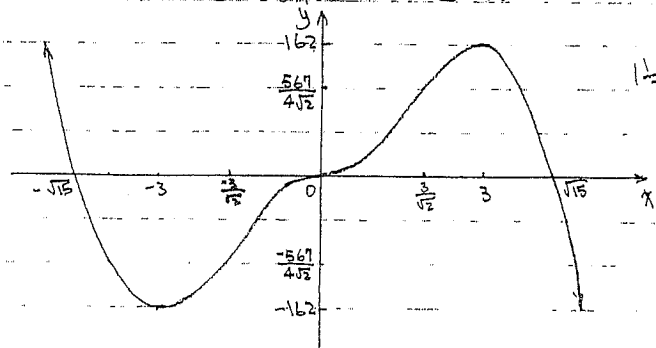
$\frac{1}{2}$ per mistake

(4)

$$ii) f(x) = -x^5 + 15x^3 = 0$$

$$x^3(-x^2 + 15) = 0$$

$$x = 0 \text{ OR } x = \pm \sqrt{15}$$



$$f''(x) = -20x^3 + 90x = 0$$

$$-10x(2x^2 - 9) = 0$$

$$x = 0 \text{ or } x = \pm \frac{3}{\sqrt{2}}$$

$$\text{When } x = \frac{3}{\sqrt{2}}, y = \frac{567}{4\sqrt{2}}$$

$$\text{and when } x = -\frac{3}{\sqrt{2}}, y = -\frac{567}{4\sqrt{2}}$$

x	$\frac{3}{\sqrt{2}}$	$\frac{3}{\sqrt{2}}$	$\frac{3}{\sqrt{2}}$
$f''(x)$	+	0	-

x	$\frac{3}{\sqrt{2}}$	$\frac{3}{\sqrt{2}}$	$\frac{3}{\sqrt{2}}$
$f''(x)$	+	0	-

$\therefore$ Points of inflexion at $(\frac{3}{\sqrt{2}}, \frac{567}{4\sqrt{2}})$ and $(-\frac{3}{\sqrt{2}}, -\frac{567}{4\sqrt{2}})$ as well.

iii) Concave down when $-\frac{3}{\sqrt{2}} < x < 0$ and $x > \frac{3}{\sqrt{2}}$

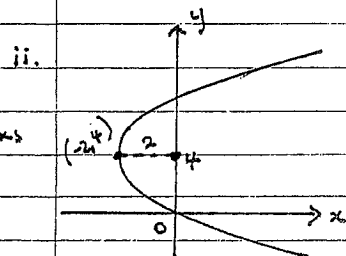
(1)

QUESTION 5

a. $(y-4)^2 = 8(x+2)$

i. Vertex is $(-2, 4)$ 1 R/W

①
MARK



$4a = 8$

$a = 2$ 1

Focus $(0, 4)$ 1

iii. y-intercept: sub $x=0$

② $(y-4)^2 = 8(0+2)$ 1

MARKS $(y-4)^2 = 16$

$y-4 = \pm 4$

$y = 0, 8$ $\frac{1}{2} + \frac{1}{2}$

b. $3x^2 - 4x - 1 = 0$

$a=3, b=-4, c=-1$

i. $\alpha + \beta = -\frac{b}{a}$

①

MARK

$= \frac{4}{3}$ 1 R/W

ii. $\alpha\beta = \frac{c}{a}$

①

MARK

$= -\frac{1}{3}$ 1 R/W

iii. $\alpha^2 + \beta^2$

② $= (\alpha + \beta)^2 - 2\alpha\beta$

MARKS $= \left(\frac{4}{3}\right)^2 - 2 \times -\frac{1}{3}$ 1

$= \frac{22}{9}$ 1

$= 2\frac{4}{9}$

c. i. $\ln 4, \ln 16, \ln 64, \dots$

①

MARK

To prove $T_2 - T_1 = T_3 - T_2$

LHS $= T_2 - T_1$

RHS $= T_3 - T_2$

$= \ln 16 - \ln 4$

$= \ln 64 - \ln 16$

$= \ln\left(\frac{16}{4}\right)$

$= \ln\left(\frac{64}{16}\right)$

$= \ln 4$

$= \ln 4$ 1

$= \text{LHS}$

As $T_2 - T_1 = T_3 - T_2$

then the sequence $\ln 4, \ln 16, \ln 64, \dots$ is arithmetic.

ii.

$a = \ln 4, d = \ln 4, n = 20$

②

MARKS

$S_n = \frac{n}{2} [2a + (n-1)d]$

$S_{20} = \frac{20}{2} [2\ln 4 + 19\ln 4]$ 1

$= 10 \times 21\ln 4$

$= 210\ln 4$ $\frac{1}{2}$

$= 210\ln 2^2$

$= 420\ln 2$ $\frac{1}{2}$

QUESTION 6

a. $v = 1 + 2t - 3t^2$

i. $\frac{dx}{dt} = 1 + 2t - 3t^2$

$x = t + t^2 - t^3 + c$

Sub $t = 0, x = 3$

$\therefore c = 3$

$x = t + t^2 - t^3 + 3$

Sub $t = 2$

$x = 2 + 2^2 - 2^3 + 3$

$x = 1$

$\therefore$ Particle is 1 cm to the right of O after 2 seconds

ii. Particle is at rest when $v = 0$

$1 + 2t - 3t^2 = 0$

$3t^2 - 2t - 1 = 0$

$(3t+1)(t-1) = 0$

$t = -\frac{1}{3}, 1$

rejected as $t \geq 0$

Particle is at rest after 1 second.

iii. $x = t + t^2 - t^3 + 3$

Sub $t = 3$

$x = 3 + 3^2 - 3^3 + 3$

$= -12$

Particle has travelled $= 1 + 12$

from part (i)

$= 13 \text{ cm}$

iv. The particle starts 3 cm to the right of O.

① It moves to the right and stops after 1 second when it is now 4 cm to the right of O.

It then moves to the left through O and carries on in that direction.

b. i. In $\triangle PMS$ and $\triangle TMR$

③ $\angle PMS = \angle TMR$ vertically opposite angles

$SM = RM$ given M is the midpoint of SR

$\angle PSM = \angle MRT$ alternate angles, $PS \parallel RT$

given PQRS is a parallelogram

$\therefore \triangle PMS \equiv \triangle TMR$ (ASA)

$\leftarrow \frac{1}{2}$

ii. $PS = RT$ corresponding sides of congruent Δ 's

② $PS \parallel RT$ given PQRS is a parallelogram then $PS \parallel QR$ and QR is produced to T.

Hence PRTS is a parallelogram as one pair of opposite sides are equal and parallel.

ALTERNATIVE METHOD

$PM = TM$ corresponding sides of congruent Δ 's

$SM = RM$ from part (i)

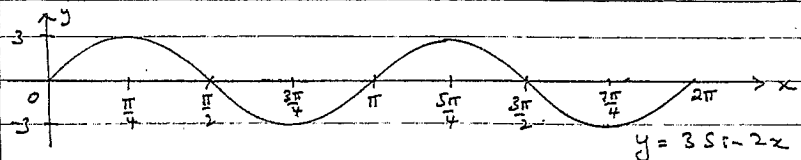
Hence PRTS is a parallelogram as both diagonals PT and SR are bisected.

QUESTION 7

2. i.

②

MARKS



$\frac{1}{2}$ mark for correct shape (no arrow-heads)

$\frac{1}{2}$ mark for correct range

$\frac{1}{2}$ mark for correct x-intercepts

$\frac{1}{2}$ mark for MAX/MIN points labelled (x-values)

ii. PERIOD = $\frac{2\pi}{2}$

①

MARK

= π R/W

iii. AMPLITUDE = 3 UNITS R/W ignore units

①

MARK

iv.

③

MARKS

Area = $\int_{\pi/4}^{\pi/2} 3 \sin 2x \, dx$ $\frac{1}{2}$

= $3 \int_{\pi/4}^{\pi/2} \sin 2x \, dx$

= $-\frac{3}{2} [\cos 2x]_{\pi/4}^{\pi/2}$ $\frac{1}{2}$

= $-\frac{3}{2} \{ \cos \pi - \cos \frac{\pi}{2} \}$ $\frac{1}{2}$

= $-\frac{3}{2} \{ -1 - 0 \}$ $\frac{1}{2}$

= $\frac{3}{2}$ units² or $1\frac{1}{2}$ units² $\leftarrow \frac{1}{2}$

Alternative Method: Area = $3 \int_0^{\pi/4} \sin 2x \, dx$

$y = 4^x$

b. i.

①

MARK

x	-1	0	1
y	$\frac{1}{4}$	1	4

ii.

②

MARKS

$\int_{-1}^1 4^x \, dx \div \frac{1}{2} \left(\frac{1}{4} + 4 + 2 \right) = 3\frac{1}{8}$

or $\int_{-1}^1 4^x \, dx \div \frac{1}{2} \left(\left(\frac{1}{4} + 1 \right) + (4 + 1) \right) = 3\frac{1}{8}$

iii.

①

MARK

$\frac{d}{dx} 4^x = \ln 4 \cdot 4^x$ or $4^x \ln 4$

①

MARK

$\beta. \int_{-1}^1 4^x \, dx$

= $\left[\frac{4^x}{\ln 4} \right]_{-1}^1$ from part d. $4^x = \frac{1}{\ln 4} \frac{d}{dx} 4^x$

= $\frac{1}{\ln 4} [4^x]_{-1}^1$

= $\frac{1}{\ln 4} \left(4 - \frac{1}{4} \right)$

= $\frac{15}{4 \ln 4}$

QUESTION 8

a.

(3)

MARKS

$$y = e^{3x} \quad (1)$$

$$\frac{dy}{dx} = 3e^{3x}$$

$$y = 4 - \frac{x}{6}$$

$$m = -\frac{1}{6}$$

$$\frac{1}{2} + \frac{1}{2}$$

$\therefore$ Gradient of Perpendicular = 6

$$\text{Wait } 3e^{3x} = 6$$

$$e^{3x} = 2$$

$$3x = \ln 2$$

$$x = \frac{1}{3} \ln 2 \quad \text{or} \quad \frac{\ln 2}{3}$$

$$\text{Sub into (1)}$$

$$y = e^{3 \cdot \frac{1}{3} \ln 2}$$

$$= e^{\ln 2}$$

$$= 2$$

$\therefore$ Coordinates of required point is $(\frac{1}{3} \ln 2, 2)$

b. i. To prove $\sin^2 x \cos x = \cos x - \cos^3 x$

(1)

MARK

$$\text{LHS} = \sin^2 x \cos x$$

$$= (1 - \cos^2 x) \cos x$$

$$= \cos x - \cos^3 x$$

$$= \text{RHS}$$

$$\therefore \sin^2 x \cos x = \cos x - \cos^3 x$$

ii.

(3)

MARKS

$$\frac{d}{dx} (\sin x - \frac{1}{3} \sin^3 x)$$

$$= \cos x - \frac{1}{3} \cdot 3 (\sin x)^2 \cdot \cos x$$

$$= \cos x - \sin^2 x \cdot \cos x$$

$$= \cos x - (\cos x - \cos^3 x) \quad \text{from part (i)}$$

$$= \cos^3 x$$

c. i.

(2)

MARKS

$$y = e^{x/2}$$

$$\text{Sub (2, e)}$$

$$e = e^{x/2}$$

$$e = e^1$$

$$\text{True}$$

$$y = \frac{e}{4} x^2$$

$$\text{Sub (2, e)}$$

$$e = \frac{e}{4} \cdot 2^2$$

$$e = e$$

$$\text{True}$$

$\therefore (2, e)$ lies on the line $y = e^{x/2}$

$e = e$
True

$\therefore (2, e)$ lies on the line $y = \frac{e}{4} x^2$

As $(2, e)$ lies on both lines and $x=2 > 0$ (1st Quadrant) then this must be the coordinates of the point A.

ii.

(3)

MARKS

$$\text{Shaded Area} = \int_0^2 (e^{x/2} - \frac{e}{4} x^2) dx$$

$$= \left[2e^{x/2} - \frac{e}{12} x^3 \right]_0^2$$

$$= (2e^1 - \frac{e}{12} \cdot 2^3) - (2e^0 - 0)$$

$$= 2e - \frac{8e}{12} - 2$$

$$= 2e - \frac{2e}{3} - 2$$

$$= e(2 - \frac{2}{3}) - 2$$

$$= \frac{4e}{3} - 2 \quad \text{units}^2$$

$$\text{Q.E.D.}$$

QUESTION 9

a. i. $\frac{dP}{dt} = kP$

②
MARKS $\Rightarrow P = P_0 e^{kt}$

Sub $P = 2P_0, t = 20$

$2P_0 = P_0 e^{20k}$

$e^{20k} = 2$

$20k = \ln 2$

$k = \frac{1}{20} \ln 2$

ii. $P = P_0 e^{(\frac{1}{20} \ln 2)t}$

②
MARKS Sub $P = 3P_0$
 $3P_0 = P_0 e^{(\frac{1}{20} \ln 2)t}$

$e^{(\frac{1}{20} \ln 2)t} = 3$

$(\frac{1}{20} \ln 2)t = \ln 3$

$t = \frac{20 \ln 3}{\ln 2}$

≈ 31.7 years

During (2006 + 31) 2037 the population will
be 3 times that which it had at the beginning of 2006.

iii. $P = P_0 e^{(\frac{1}{20} \ln 2)t}$

② Sub $P_0 = 20, t = 50$

MARKS

$P = 20 e^{(\frac{1}{20} \ln 2)50}$

$= 20 e^{\frac{5}{2} \ln 2}$

$= 113.137085$ million

or 113 137 085

REFER TO NEXT PAGE FOR ALTERNATIVE SOLUTION

(b) i. 24% p.a. = 2% per month

① $A_1 = Px1.02 - 400$

MARK

$A_2 = A_1 \times 1.02 - 400$

$= (Px1.02 - 400) \times 1.02 - 400$

$\therefore A_2 = Px1.02^2 - 400(1 + 1.02)$

ii. $A_3 = A_2 \times 1.02 - 400$

$= Px1.02^3 - 400(1 + 1.02 + 1.02^2)$

③

MARKS

Similarly

$A_n = Px1.02^n - 400(1 + 1.02 + 1.02^2 + \dots + 1.02^{n-1})$

This is a Geometric Series

$a=1, r=1.02, n=n$

$S_n = a \frac{(r^n - 1)}{r - 1}$

$= \frac{1(1.02^n - 1)}{1.02 - 1}$

$\therefore A_n = Px1.02^n - 400 \frac{(1.02^n - 1)}{0.02}$

$= Px1.02^n - 20000(1.02^n - 1)$

Q.E.D.

$$\text{iii. } n = 5 \times 12 = 60$$

MARKS

$$A_{60} = P \times 1.02^{60} - 20000(1.02^{60} - 1) \quad \frac{1}{2}$$

$$\text{But } A_{60} = 0 \quad [\text{LOAN IS REPAYED}] \quad \frac{1}{2}$$

$$\therefore P \times 1.02^{60} - 20000(1.02^{60} - 1) = 0$$

$$P = \frac{20000(1.02^{60} - 1)}{1.02^{60}} \quad \frac{1}{2}$$

$$= \$13\,904.35 \quad \frac{1}{2}$$

Sarah is able to borrow \$13 904 to the nearest dollar.

Part (a) (iii) ALTERNATIVE SOLUTION

In 2006, population P_0

2010, population 2 Million, $\therefore t = 4$

$$20 = P_0 e^{4k}$$

$$P_0 = \frac{20}{e^{(\frac{1}{50} \ln 4) \times 4}} = 17.41101127 \text{ Million} \quad (1)$$

From 2006 to 2060 — 54 years. $(\frac{1}{2})$

$$\begin{aligned} \text{a. } P &= P_0 e^{(\frac{1}{50} \ln 4) \times 54} \\ &= 113\,137\,085 \end{aligned} \quad (\frac{1}{2})$$

QUESTION 10

$$\text{a. } \frac{d}{dx} \ln(x + \sqrt{x^2 + a^2})$$

(3)

MARKS

$$= \frac{d}{dx} \ln [x + (x^2 + a^2)^{1/2}]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \times \left[1 + \frac{1}{2} (x^2 + a^2)^{-1/2} \cdot 2x \right] \quad \frac{1}{2} + \frac{1}{2} +$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \times \left[1 + \frac{x}{\sqrt{x^2 + a^2}} \right]$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \times \frac{\sqrt{x^2 + a^2} + x}{\sqrt{x^2 + a^2}}$$

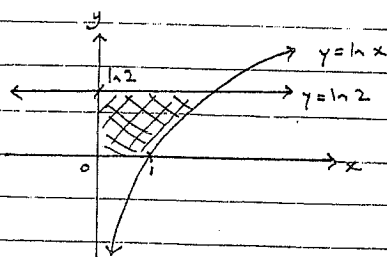
$$= \frac{1}{\sqrt{x^2 + a^2}} \quad \frac{1}{2}$$

$$\begin{aligned} \int \frac{1}{2\sqrt{x^2 + a^2}} dx &= \frac{1}{2} \int \frac{1}{\sqrt{x^2 + a^2}} dx \\ &= \frac{1}{2} \int \left[\frac{d}{dx} \ln(x + \sqrt{x^2 + a^2}) \right] dx \\ &= \frac{1}{2} \ln(x + \sqrt{x^2 + a^2}) + C \quad \frac{1}{2} + \frac{1}{2} \text{ for } +C \end{aligned}$$

b.

(3)

MARKS



$$y = \ln x$$

$$x = e^y$$

$$x^2 = e^{2y}$$

$$V = \pi \int_{-1}^0 x^2 dy$$

$$= \pi \int_{-1}^0 e^{2y} dy$$

$$\therefore V = \frac{\pi}{2} [e^{2y}]_0^{\frac{1}{2}}$$

$$= \frac{\pi}{2} \{ e^{2 \ln 2} - e^0 \}$$

$$= \frac{\pi}{2} \{ e^{\ln 4} - 1 \}$$

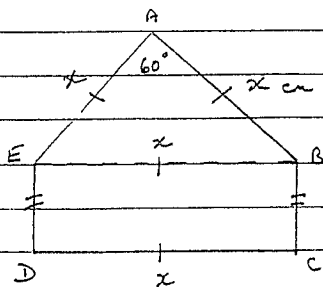
$$= \frac{\pi}{2} (4 - 1)$$

$$= \frac{3\pi}{2} \text{ units}^3$$

[N.B. without π then
2 marks in total]

C. i.

①
MARK



ED = BC opposite sides of a rectangle BCDE

$$\text{Perimeter} = AB + BC + CD + DE + EA$$

AB = AE = x = EB Equal sides of Equilateral triangle ABE

EB = DC = x Opposite sides of rectangle BCDE

$$P = x + BC + x + BC + x$$

$$P = 3x + 2BC$$

$$2BC = P - 3x$$

$$BC = \frac{P - 3x}{2}$$

ii.

Area = Area of $\triangle ABE$ + Area of Rectangle BCDE

②

MARKS

$$= \frac{1}{2} ab \sin C + LB$$

$$= \frac{1}{2} \cdot x \cdot x \cdot \sin 60^\circ + x \cdot \frac{P - 3x}{2}$$

$$= \frac{1}{2} x^2 \cdot \frac{\sqrt{3}}{2} + \frac{Px - 3x^2}{2}$$

$$= \frac{1}{4} [\sqrt{3}x^2 + 2(Px - 3x^2)]$$

$$= \frac{1}{4} [\sqrt{3}x^2 + 2Px - 6x^2]$$

$$= \frac{1}{4} [2Px - x^2(6 - \sqrt{3})]$$

$$\therefore A = \frac{1}{4} [2Px - (6 - \sqrt{3})x^2] \quad \text{Q.E.D.}$$

iii.

Maximum Area occurs when $\frac{dA}{dx} = 0$

③

MARKS

$$\frac{dA}{dx} = \frac{1}{4} [2P - 2(6 - \sqrt{3})x]$$

$$= \frac{1}{2} [P - (6 - \sqrt{3})x]$$

$$\text{Hence want } \frac{1}{2} [P - (6 - \sqrt{3})x] = 0$$

$$P - (6 - \sqrt{3})x = 0$$

$$P = (6 - \sqrt{3})x$$

$$\frac{P}{x} = 6 - \sqrt{3}$$

$$\frac{d^2A}{dx^2} = -\frac{1}{2} (6 - \sqrt{3})$$

$\frac{d^2A}{dx^2} < 0$ for all values of x $\therefore$ Max Area occurs when $\frac{P}{x} = 6 - \sqrt{3}$.